



Fix the Stack

Quick facilitation notes & answer key

Goal

Children debug a block program: they run it, see that Bit misses the star, find the one block that points the wrong way, and fix it so the program works. Run → find the bug → fix → run again is the debugging cycle, and it closes the Intro Block Coding unit.

Kindergarten frame (Ontario)

Problem Solving and Innovating — children test a sequence, identify where it goes wrong, and correct it.

No reading required of the child: read each prompt aloud; the child writes a box number, circles a block, and draws one arrow.

Materials

The printed worksheet, a pencil, and a blue crayon. Optional: floor boxes 1-5 with a star on box 4 so children can walk the buggy program and feel Bit go the wrong way.

Run it (about 15 min)

1. MODEL: run the buggy program together — ➡ to box 2, ⬅ back to box 1, ➡ to box 2. 'Bit is on box 2, not the star. Something is wrong.'
2. GUIDED: Exercise 1 — Bit lands on box 2 (not the star). Exercise 2 — the middle ⬅ block sent Bit backward; circle it.
3. YOU TRY: Exercise 3 — the fixed block should be ➡; the program becomes ➡ ➡ ➡ and Bit reaches box 4, the star.
4. Connect: 'Coders do this all the time — they run a program, find the wrong block, and fix it. That is debugging.'

Answer key (modelled by the run-gate)

Ex 1 — the buggy program (➡ ⬅ ➡) from box 1 lands Bit on box 2. Not the star.

Ex 2 — the wrong block is the middle ⬅ (it sends Bit back one box).

Ex 3 — fix it to ➡: the program becomes ➡ ➡ ➡ and Bit lands on box 4, the star.

Watch for & differentiate

Common slip: children blame the first or last block — have them run step by step and watch exactly where Bit turns back.

Anchor: the wrong block is the one that moves Bit AWAY from the star.

Simplify: walk the buggy program on the floor so the backward step is felt, then fix and re-walk.

Extend: ask children to invent their own one-block bug for a partner to find and fix.

Movement option: a child acts as Bit and walks the buggy program, freezing on box 2; the class calls out the fix.

Success looks like

The child writes box 2 for the buggy run, circles the middle ⬅ block, draws ➡ to fix it, and runs the fixed program to land Bit on box 4, the star.